

Ricardo Brutus

Brockton, MA | ricardobrutus.rb@gmail.com | 774-381-2946 | www.ricardobrutus.com | linkedin.com/in/ricardobrutus

SUMMARY

Computer Science student at UMass Boston (Expected May 2027) with hands-on experience building and shipping full-stack and iOS applications. Proficient in Python, JavaScript, Swift, React, Flask, and AWS. Experience building LLM agent systems and real-time production apps with Firebase, SwiftUI, and REST APIs, plus leading technical instruction for 40+ students.

EDUCATION

University of Massachusetts Boston Boston, MA

B.S. Computer Science | GPA: 3.4/4.0 | **Expected May 2027**

Coursework: Computer Architecture & Organization, Data Structures & Algorithms, Systems Programming, Quantum Computing

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, Swift

Frameworks & Tools: React, Flask, SwiftUI, Node.js, HTML/CSS, Git, Docker, Firebase, AWS (S3, EC2, Lambda)

Databases & Testing: SQL, PostgreSQL, Jest, Pytest

APIs & Concepts: REST APIs, LLM Agents & Function Calling, RAG, Agile/Scrum, CI/CD

Leadership/Experience

Computer Science Club — UMass Boston Boston, MA

Member & Peer Tutor 2025 – present

- Tutored fellow CS students in core topics including data structures, algorithms, and systems programming.
- Participated in coding workshops and Leetcode-focused interview prep sessions, sharpening problem-solving skills in algorithms and time complexity.

Teaching Assistant — UMass Boston Boston, MA

CS 110 intro to programming Fall 2024

- Led weekly lab sessions/office hours for 40+ students, debugging code and explaining core Python concepts (loops, functions, data structures)
- Graded assignments and provided feedback on 40+ submissions per week

HackUmass — Amherst, MA

AI Hackathon 2024 – 2024

- Competed in a team-based AI hackathon, placing in the top 30 out of all participating teams

PROJECTS

Earworm — Ai Song Recognition & Recommendation Agent

github.com/rickybr34/earworm

Python, Node.js, React, Groq (LLM function-calling), PyTorch, Essentia

- Architected an LLM agent (Groq/Llama, function-calling) that autonomously orchestrates song recognition, audio analysis, and recommendation tools based on conversation context, replacing a fixed backend pipeline with model-driven tool selection
- Built a query-by-humming recognition system using pitch-contour extraction and Dynamic Time Warping, identifying songs from hummed/sung audio without exact-match fingerprinting
- Trained a PyTorch neural network on audio features (tempo, energy, danceability) to learn genre/mood embeddings, replacing hand-coded similarity search with a learned representation
- Implemented persistent taste-profiling and retrieval-augmented generation (RAG) over song lyrics/context, enabling multi-turn personalization grounded in retrieved data

MunchMatch — Group Restaurant Decision App

github.com/rickybr34/MunchMatch-showcase

Swift, SwiftUI, Firebase, Google Places API

- Building a real-time iOS app where groups create shared rooms, set food preferences, and swipe Tinder-style on nearby restaurants via Google Places API to reach a group consensus.
- Implemented Firebase Realtime Database for live room sync, Firebase Authentication for user identity, and push notifications for session updates.
- Targeting TestFlight release with crash reporting and analytics integrated for stability ahead of App Store submission.

Luvly — Couple Connection App

github.com/rickybr34/Luvly-Demo

Swift, SwiftUI, Combine, MVVM

- Designing and building a full-featured iOS relationship app with real-time partner notifications, interactive quiz games (Never Have I Ever, This or That, Who's More Likely To), daily questions, and anniversary countdown.
- Designed using MVVM pattern with Combine and ObservableObject for reactive state management across shared partner data.
- Implemented a topic-based quiz system with smart filtering, progress tracking, and a multi-step onboarding flow with secure 6-digit partner pairing codes.